

Yashvardhan Singh

AI & Data Systems Engineer · LLM Infrastructure · Edge ML · Data Pipelines

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PROFESSIONAL SUMMARY

AI systems engineer specialising in the latency economics of edge ML inference. Published at ICCCN 2026 on vision encoder optimisation for edge VLLMs; benchmarked 8 architectures across 4 hardware platforms with TTFT ranging from 1.6 ms to 98.0 ms. Introduced the Cache Cliff phenomenon and Sensory Bottleneck Hypothesis. Built and shipped production data pipelines at Medikabazaar and a live LLM orchestration engine serving real users.

RESEARCH

On-Device AI - Vision Encoder Optimization for Edge Visual LLMs

Accepted · ICCCN 2026, UK

EfficientNet (B0-B4) · EfficientFormer · MobileViT · Raspberry Pi 5 · Google Coral TPU · NVIDIA Jetson Orin Nano · TensorRT · TFLite

- Measured vision encoders as the 70-85% TTFT bottleneck in edge VLLM deployments; benchmarked 8 architectures across 4 hardware platforms, results ranging from 1.6 ms (EfficientFormer-L1) to 98.0 ms (ViT-Base) on identical hardware.
- Introduced the Cache Cliff phenomenon (non-linear latency jump when model size exceeds L2/L3 NPU cache) and the Sensory Bottleneck Hypothesis; produced a hardware-aware encoder selection matrix across three latency regimes: real-time (15 ms), interactive (100 ms), and batch (500 ms).

A Systematic Review of Deep Learning Approaches for Motor Imagery EEG-Driven BCIs

Peer-reviewed at IIT Kharagpur · IEEE-format · 100+ studies synthesised · 17 architectures benchmarked on BCI Competition IV-2a

PROFESSIONAL EXPERIENCE

Technology Intern - Medikabazaar

June 2025 - August 2025

B2B FinTech & Procurement · Gurugram, India · Python · AWS (EC2, S3, RDS) · Kafka · Redis · ML inference

- Optimized an AWS OCR ML inference pipeline for warehouse document digitization; improved model extraction accuracy, structured output capture, and reduced manual correction overhead at scale.
- Replaced a legacy batch financial workflow with an event-driven Kafka pipeline; reduced end-to-end credit note processing latency from 3 days to under 10 minutes across 400+ monthly requests.
- Delivered data validation and standardisation workflows across the seller entity database, improving downstream data quality and reliability across 28 states.

PROJECTS

DiagFlo - Visualisation Engine : Natural Language to Artifacts

diagflo.yashvardhan.dev

LangChain · MCP Server · LLM orchestration · Gemini API · Cloudflare

- LLM engine converts plain-language inputs into executable DSLs across 20+ technical formats via in-context learning and multi-modal prompting; deployed and serving ~50 weekly product users.
- Fault-tolerant AI orchestration layer with MCP server integration, exponential backoff, and dual-renderer routing sustains uptime across LLM client failures without manual intervention.

GetCred - Alternative Credit Intelligence for India's New-To-Credit Economy

getcred.yashvardhan.dev

Python · XGBoost · SQLite · PySpark · SHAP · Docker

- Trained a bureau-agnostic ML scoring model on transaction velocity, merchant category codes, and income volatility signals; targeted the 120M+ unbanked Indian demographic underserved by static bureau scoring.
- Privacy-first data pipeline using the Account Aggregator framework processes 2M+ simulated monthly transactions at sub-2-minute inference latency; PII stripped at the edge before ingestion.

TECHNICAL SKILLS

AI & LLM Systems: LangChain · LLM orchestration · MCP Server · Prompt engineering · Structured outputs · TensorRT · TFLite · Edge ML inference · XGBoost · Feature engineering

Data Engineering: SQL (MySQL, PostgreSQL) · query optimization · ETL · Kafka · PySpark · Redis · Data modeling · Airflow · Orchestration · Data pipelines · Linux server administration (Nginx, Systemctl, Bare-metal EPYC)

Cloud & Infra: AWS (EC2, S3, RDS) · GCP (Compute Engine, Cloud run) · Docker · CI/CD · Bash · Cloudflare · Vercel · Jenkins

Languages & Tools: Python · SQL · Git · Postman

EDUCATION

B.Tech Information Technology (Honors), KIIT University, Bhubaneswar

CGPA: 8.38/10 · 2022 - 2026

School of Computer Engineering · Relevant Coursework: Cloud Computing · Big Data · Artificial Intelligence · Computer Networking · Data Analytics

CERTIFICATIONS

IBM Generative AI: Prompt Engineering [Coursera](#) & **Applications:** [Coursera](#)

LEADERSHIP

Technical Lead - Microsoft Learn Student Ambassadors, KIIT

March 2023 - May 2025

3,000+ member community · Bhubaneswar · mksakiit.com

- Led the largest technical event in Odisha, 3,500+ participants across AI, ML, and data tooling; designed and delivered a hands-on workshop for 400+ attendees, recognised as event of the year by the university dean.